

EXPERIMENT – 16 - SERIAL INTERRUPT USING UART (KEYBOARD INPUT) IN KEIL

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PROGRAM

```
#include <reg51.h>
#include <stdio.h>

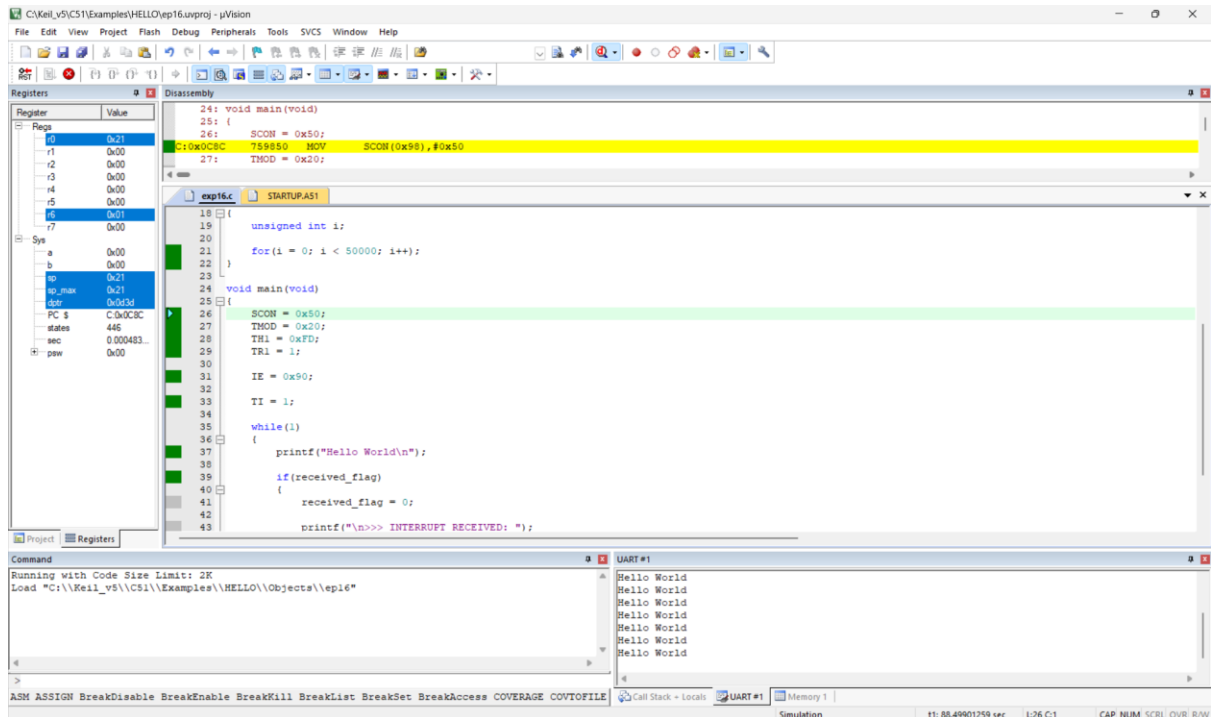
void serial_ISR(void) interrupt 4 {
    char ch;
    if (RI)
    {
        RI = 0; // Clear receive interrupt flag
        ch = SBUF; // Read received character
        printf("\n>>> INTERRUPT RECEIVED: ");
        printf("%c\n", ch);
    } }

void delay(void) {
    unsigned int i;
    for(i = 0; i < 50000; i++); // Delay for visibility }

void main(void) {
    SCON = 0x50; // UART mode 1, 8-bit, REN enabled
    TMOD = 0x20; // Timer1 mode 2 (auto-reload)
    TH1 = 0xFD; // Baud rate 9600
    TR1 = 1; // Start Timer1
    IE = 0x90; // EA = 1, ES = 1 (enable serial interrupt)
    TI = 1; // Enable printf
    while(1) {
        printf("Hello World\n");
        delay();
    } }
```

OUTPUT

Displaying a Message:



During interruption:

